HFD23 (JRC-23F) SUBMINIATURE SIGNAL RELAY

c UL us
File No.:E133481

CQC
File No.:CQC02001001948

Features

- Max.2A switching capability
- High sensitive: 150mW
- 1 Form C configuration
- Wash tight type available
- Environmental friendly product (RoHS compliant)
- Outline Dimensions: (12.5 x 7.5 x 10.0) mm

CONTACT DATA

Contact arrangement	1C
Contact resistance	100m Ω(at 0.1A 6VDC)
Contact material	AgNi +Au plated
Contact rating (Res. load)	0.5A 125VAC / 1A 30VDC
Max. switching voltage	125VAC / 60VDC
Max. switching current	2A
Max. switching power	62.5VA / 30W
Min. permissible load	1mA 5V
Mechanical endurance	$1x10^7$ OPS (300OPS/min.)
Electrical endurance	$1x10^5$ OPS (30OPS/min.)

CHARACTERISTICS

Insulation resistance		1000M Ω(at 500VDC)
Dielectric strength	Between coil & contacts	1000VAC 1min
	Between open contacts	400VAC 1min
Operate time (at nomi. volt.)		5ms max.
Release time (at nomi. volt.)		5ms max.
Bounce time (at nomi.volt.)		5ms max.
Temperature rise (at nomi.volt.)		65K max.
Shock resistance		100m/s² (10g)
Vibration resistance		10Hz to 55Hz 3.3mm DA
Humidity		35% to 85% RH
Ambient temperature		-30˚C to 70˚C
Unit weight		Approx. 2.2g
Termination		PCB (DIP)
Construction		Wash tight

Notes: The data shown above are initial values.

COIL

Coil power	Sensitive:150mW; Standard: 200mW

COIL DATA at 23˚C

Standard Type (200mW)

Nominal Voltage VDC	Pick-up Voltage VDC	Drop-out Voltage VDC	Max. Allowable Voltage VDC	Coil Resistance Ω
1.5	1.20	0.15	2.25	11.3 x (1±10%)
3	2.40	0.30	4.5	45 x (1±10%)
5	4.00	0.50	7.5	125 x (1±10%)
6	4.80	0.60	9.0	180 x (1±10%)
9	7.20	0.90	13.5	405 x (1±10%)
12	9.60	1.20	18.0	720 x (1±10%)
24	19.20	2.40	36.0	2880 x (1±15%)

Sensitive Type (150mW)

Nominal Voltage VDC	Pick-up Voltage VDC	Drop-out Voltage VDC	Max. Allowable Voltage VDC	Coil Resistance Ω
1.5	1.20	0.15	2.25	15 x (1±10%)
3	2.40	0.30	4.5	60 x (1±10%)
5	4.00	0.50	7.5	167 x (1±10%)
6	4.80	0.60	9.0	240 x (1±10%)
9	7.20	0.90	13.5	540 x (1±10%)
12	9.60	1.20	18.0	960 x (1±10%)
24	19.20	2.40	36.0	3840 x (1±15%)

SAFETY APPROVAL RATINGS

UL&CUR	1.0A 30VDC 0.3A 60VDC 0.5A 125VAC

Notes: Only some typical ratings are listed above. If more details are required, please contact us.

ORDERING INFORMATION

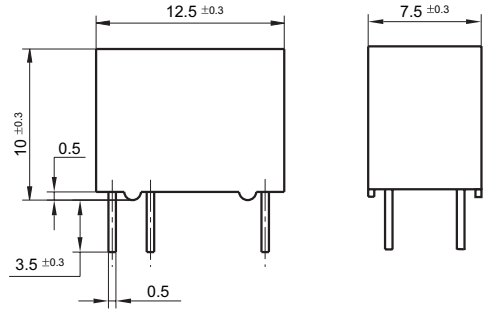
	HFD23 /	012	-1Z	S	(XXX)
Type [1]	HFD23 JRC-23F (Old type)				
Coil voltage	1.5, 3, 5, 6, 9, 12, 24VDC				
Contact arrangement	**1Z:** 1 Form C				
Coil power	**S:** Sensitive type (150mW)	**P:** Standard type (200mW)			
Customer special code [2]	Only for special requirements, e.g. (555) stands for RoHS compliant				

Notes: 1) We have now gradually updated our ordering information. We suggest new type should be selected. If necessary, old type can be kept for some period for the old customers.
2) HFD23 is an environmental friendly product. Please mark a special code (555) when ordering.

OUTLINE DIMENSIONS, WIRING DIAGRAM AND PC BOARD LAYOUT

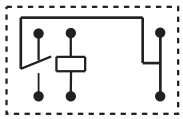
Unit: mm

Outline Dimensions



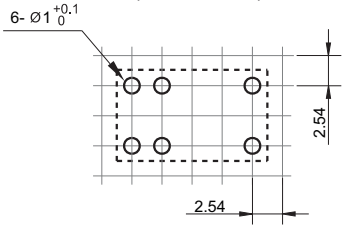
Wiring Diagram

(Bottom view)



PCB Layout

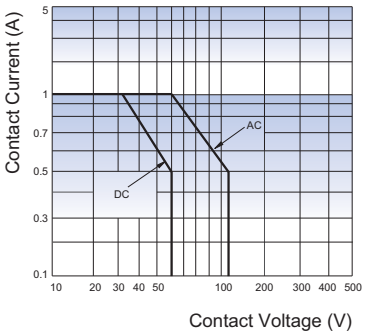
(Bottom view)



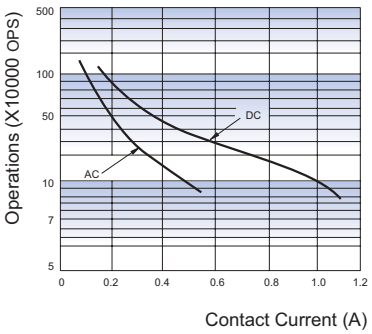
Remark: 1) In case of no tolerance shown in outline dimension: outline dimension ≤1mm, tolerance should be ±0.2mm; outline dimension >1mm and ≤5mm, tolerance should be ±0.3mm; outline dimension >5mm, tolerance should be ±0.4mm.
2) The tolerance without indicating for PCB layout is always ±0.1mm.
3) The width of the gridding is 2.54mm.

CHARACTERISTIC CURVES

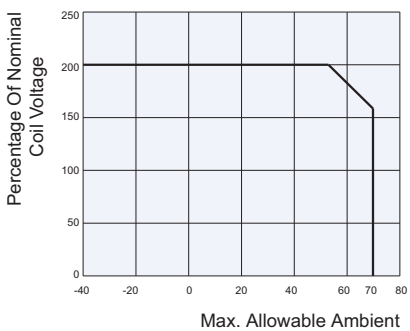
MAXIMUM SWITCHING POWER



ENDURANCE CURVE



MAX.ALLOWABLE AMBIENT TEMPERATURE CURVE



Disclaimer

This datasheet is for the customers' reference. All the specifications are subject to change without notice.
We could not evaluate all the performance and all the parameters for every possible application. Thus the user should be in a right position to choose the suitable product for their own application. If there is any query, please contact Hongfa for the technical service. However, it is the user's responsibility to determine which product should be used only.

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